Artificial Intelligence

Deep learning 5.3 - *Foundation models*

Master in Artificial Intelligence & Computational engineering

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- D1.08 Ada Byron - Tu. 12:00-14:00, Th. 10:00-14:00

Please, recommended to email to book/confirm office hour slots (office hours may be in person or online).

Some reminders

- LAB 4 submission: this Wednesday
- **LAB 5: tomorrow**
  - If possible review the prior work to be prepared
    - It's short, to submit right after the session or next day. We'll evaluate it on Thursday with an interview.
- **Final Exam:** short questions (everyone, around 1h) + Global exam about labs (if not submitted in the continuous evaluation)

Today

- **So far:** *Supervised Deep learning - CNN & Transformers*
- **Next: Foundation models & …**
    - Other supervision strategies
        - Unsupervised
        - RL
    - Generative models

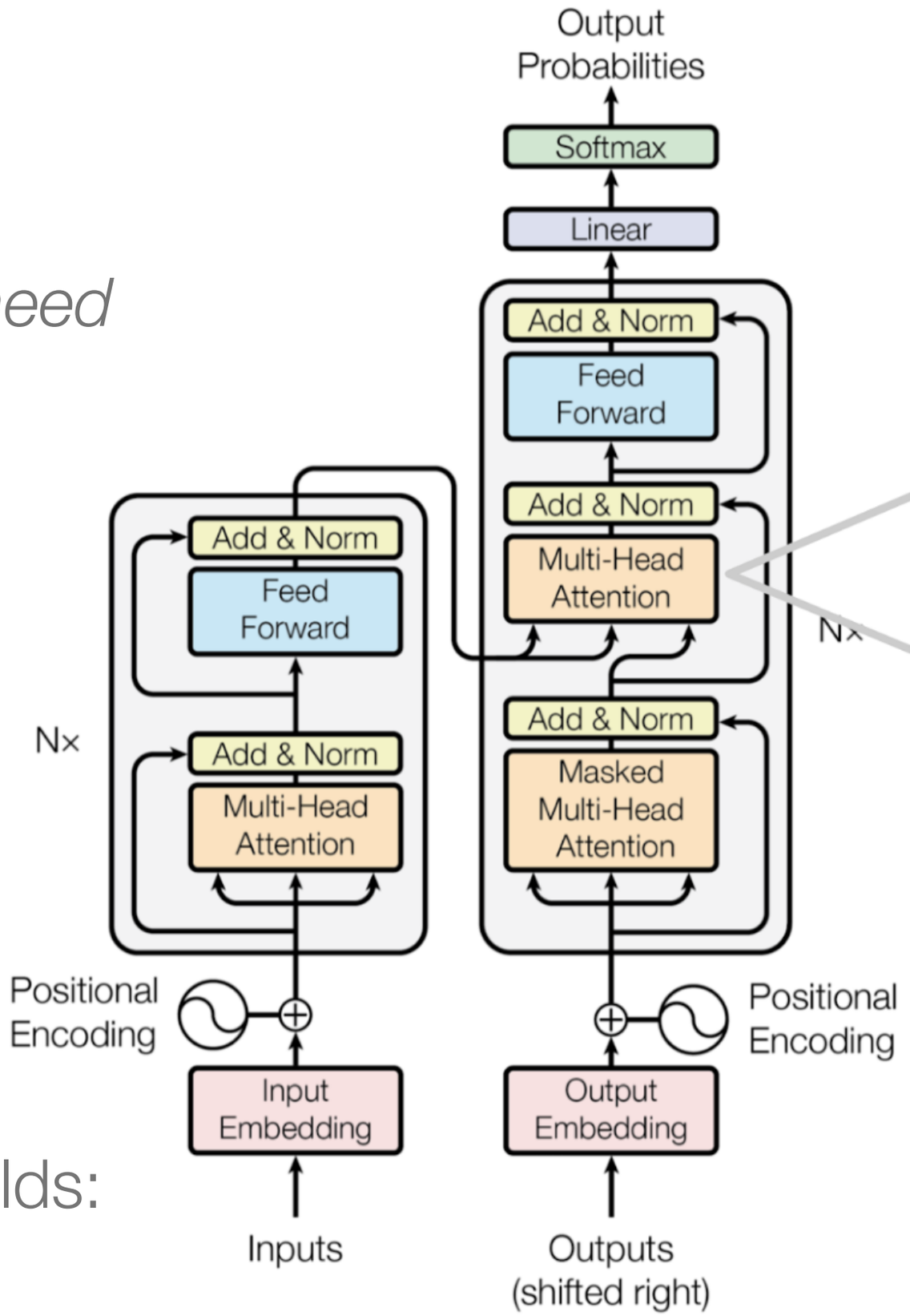
Transformers

Transformers: *Attention is all you need*

- Based on *self-attention:*
  similar to FC layer,
  but instead of fixed weights,
  they are adapted
  depending on the input

- Usually multi-head: multiple
  sets of learned weights

- Dominant paradigm in many fields:
  NLP, Computer Vision, …



A. Vaswani et al. “Attention Is All You Need”. In: NIPS. 2017.

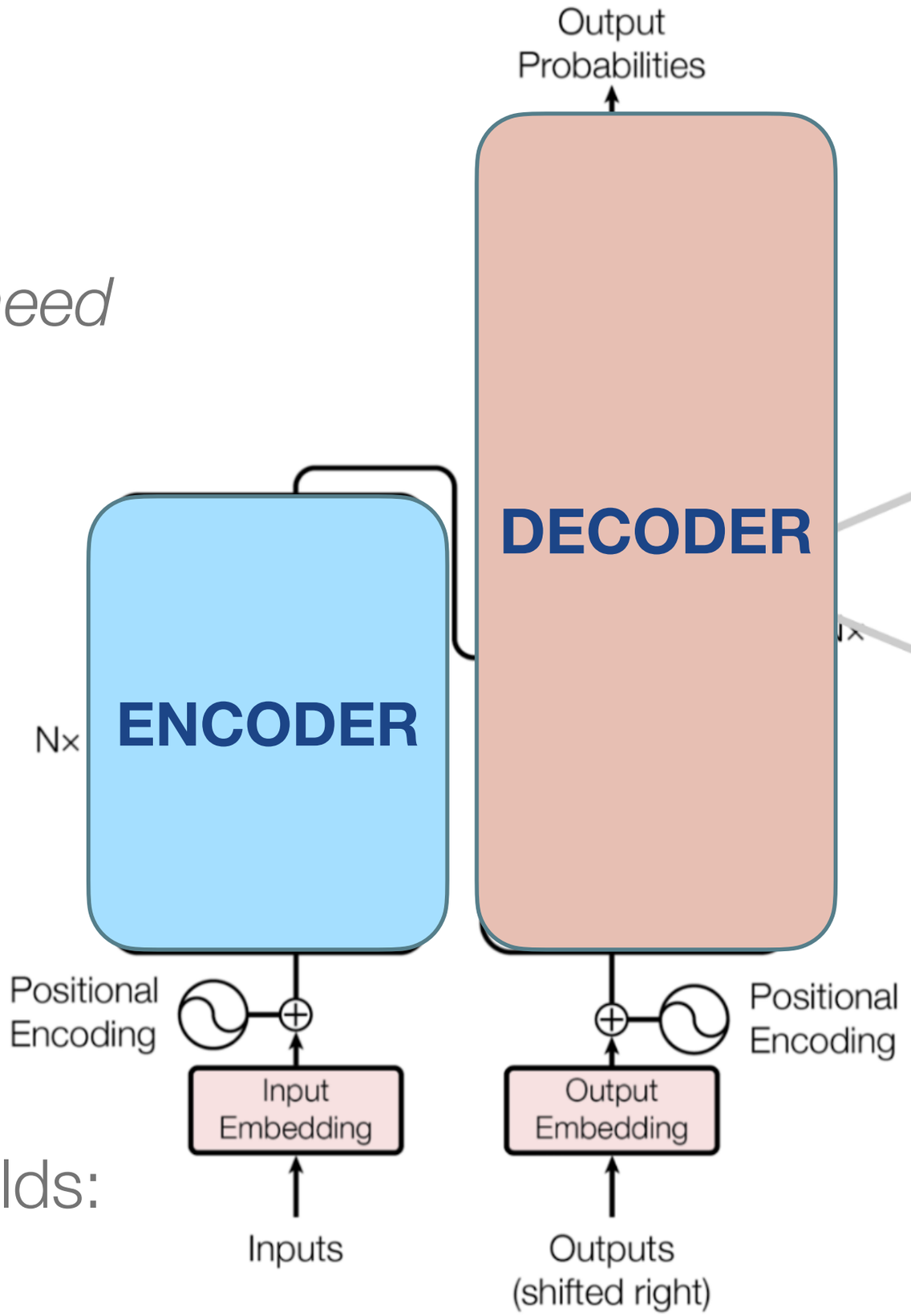
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Transformers

Encoder models (e.g. **BERT**)

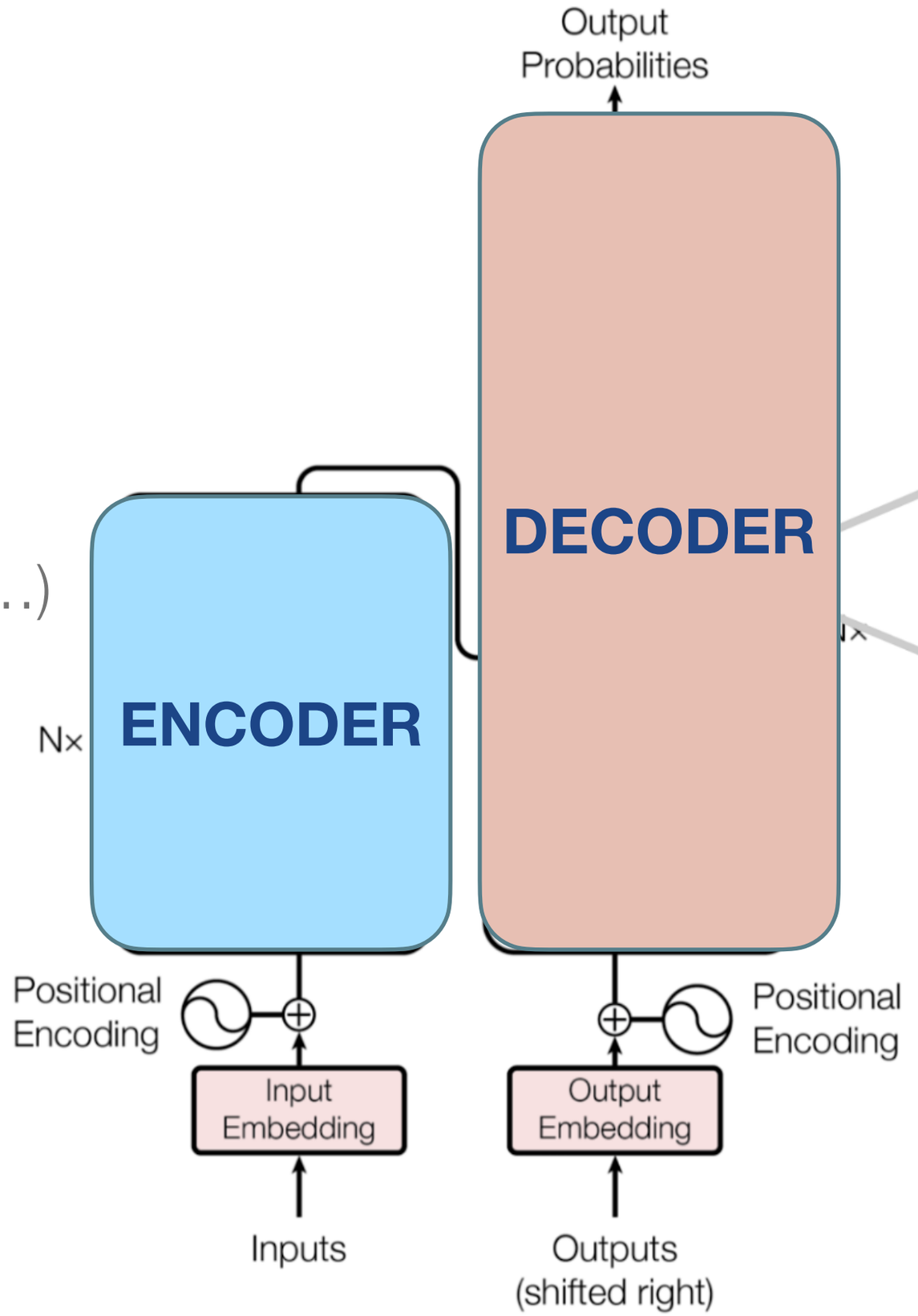
- Embeddings from input tokens

Decoder models (e.g. **GPT, Gemini**, …)

- Left to right autoregressive generation

LAB 5!

- Decoder won't know the future during inference —> attention can't use future tokens during training (use mask attention to ignore them)



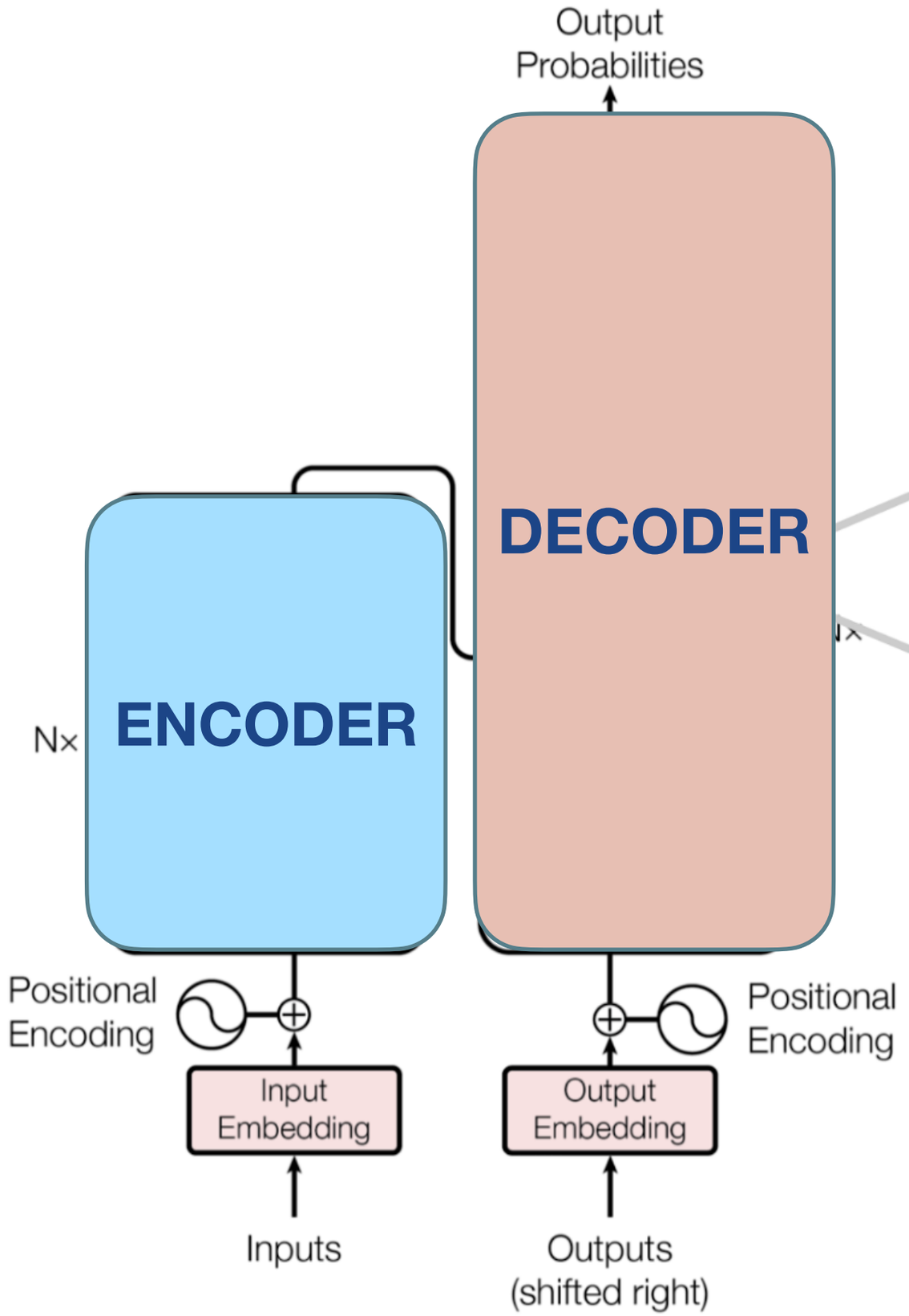
A. Vaswani et al. "Attention Is All You Need". In: NIPS. 2017.

Transformers

Encoder-Decoder models
(e.g. **T5, Whisper**)

- Seq2seq models (translation, transcription, …)

- *Cross attention:*
  Queries come from the previous decoder layer,
  but Keys and Values
  come from encoder output



A. Vaswani et al. “Attention Is All You Need”. In: NIPS. 2017.

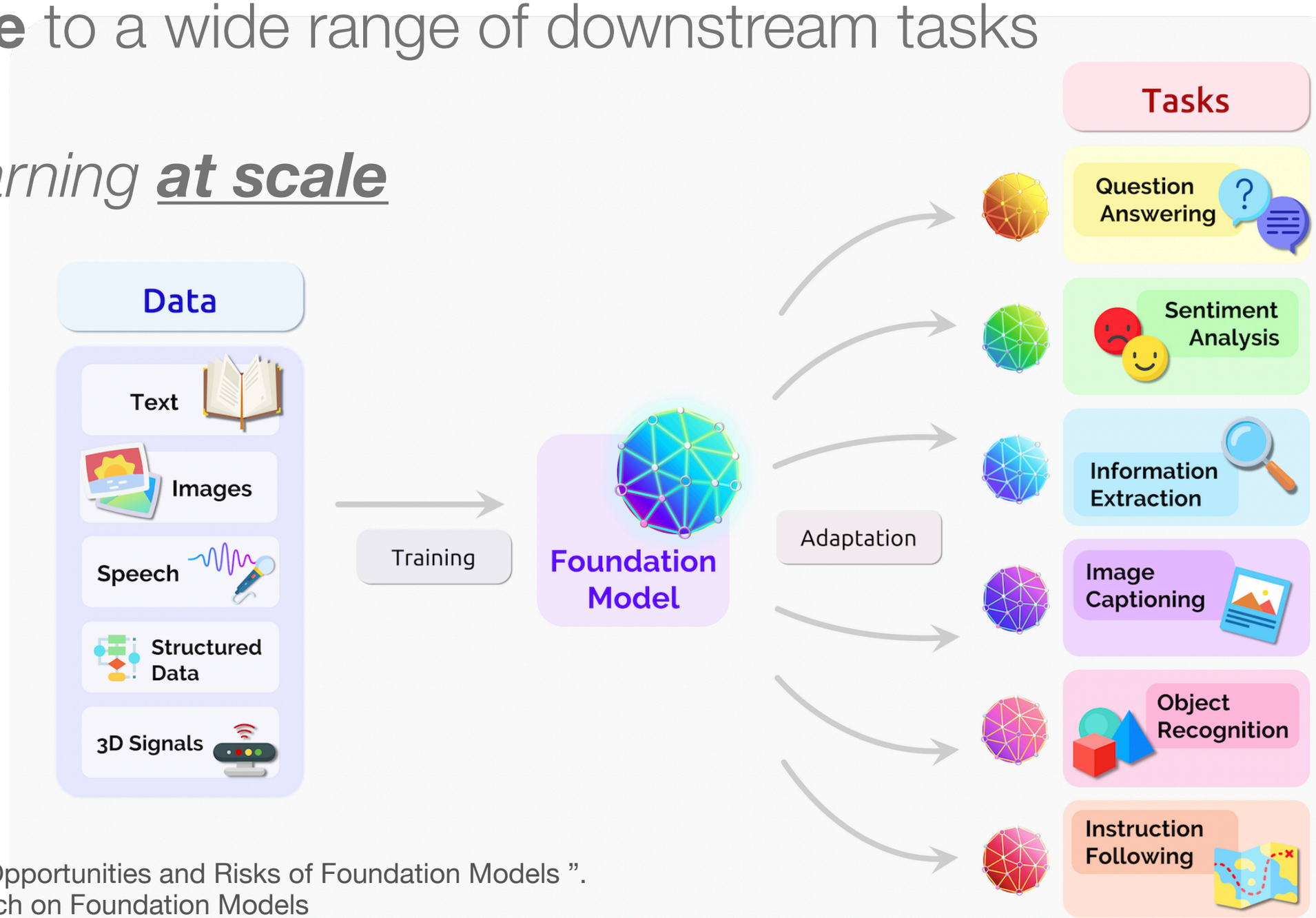
Foundation models

Transformers &
Foundation models?

Foundation models

- Trained on broad **very (very) large scale** datasets and **adaptable** to a wide range of downstream tasks

Transfer learning ***at scale***

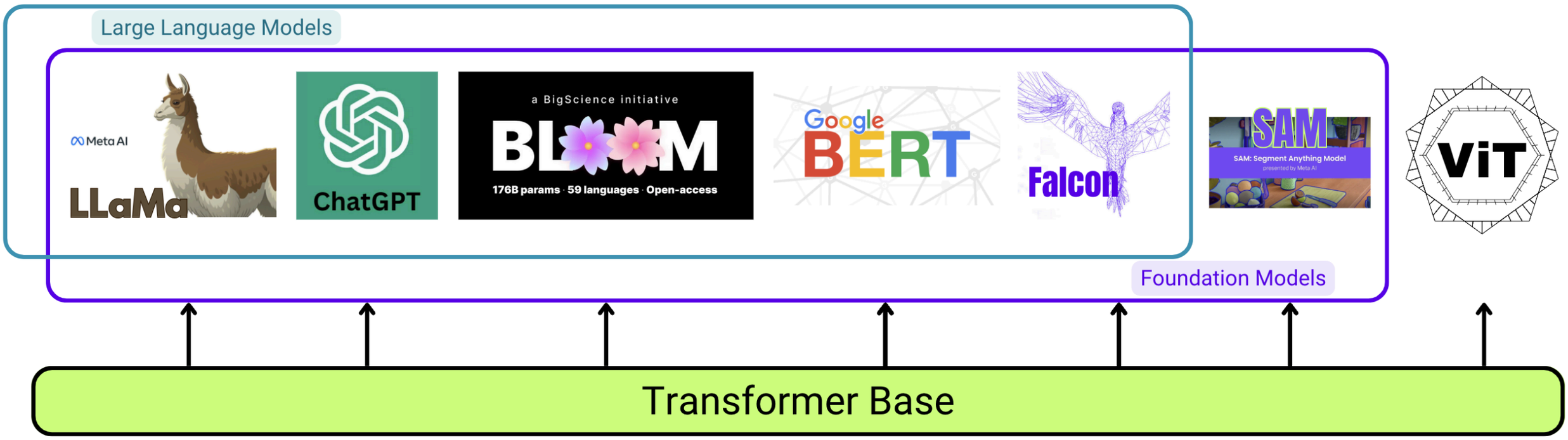


R. Bommasani et al. “On the Opportunities and Risks of Foundation Models ”. arXiv. 2022. Center for Research on Foundation Models

Foundation models

- Trained on broad **very (very) large scale** datasets and **adaptable** to a wide range of downstream tasks

Transfer learning ***<u>at scale</u>***



https://www.comet.com/site/blog/explainable-ai-for-transformers/

Foundation models

Pros: plenty application domains & Impressive results

Cons: Very (VERY) resource demanding

- GPT-3 has 175B params. *Frontier models* unknown - believed to be Mixture-of-Experts

- DeepSeek-V3: 671B params. About 37B active per token Kimi K2 about 1T total, about 32B active per token)

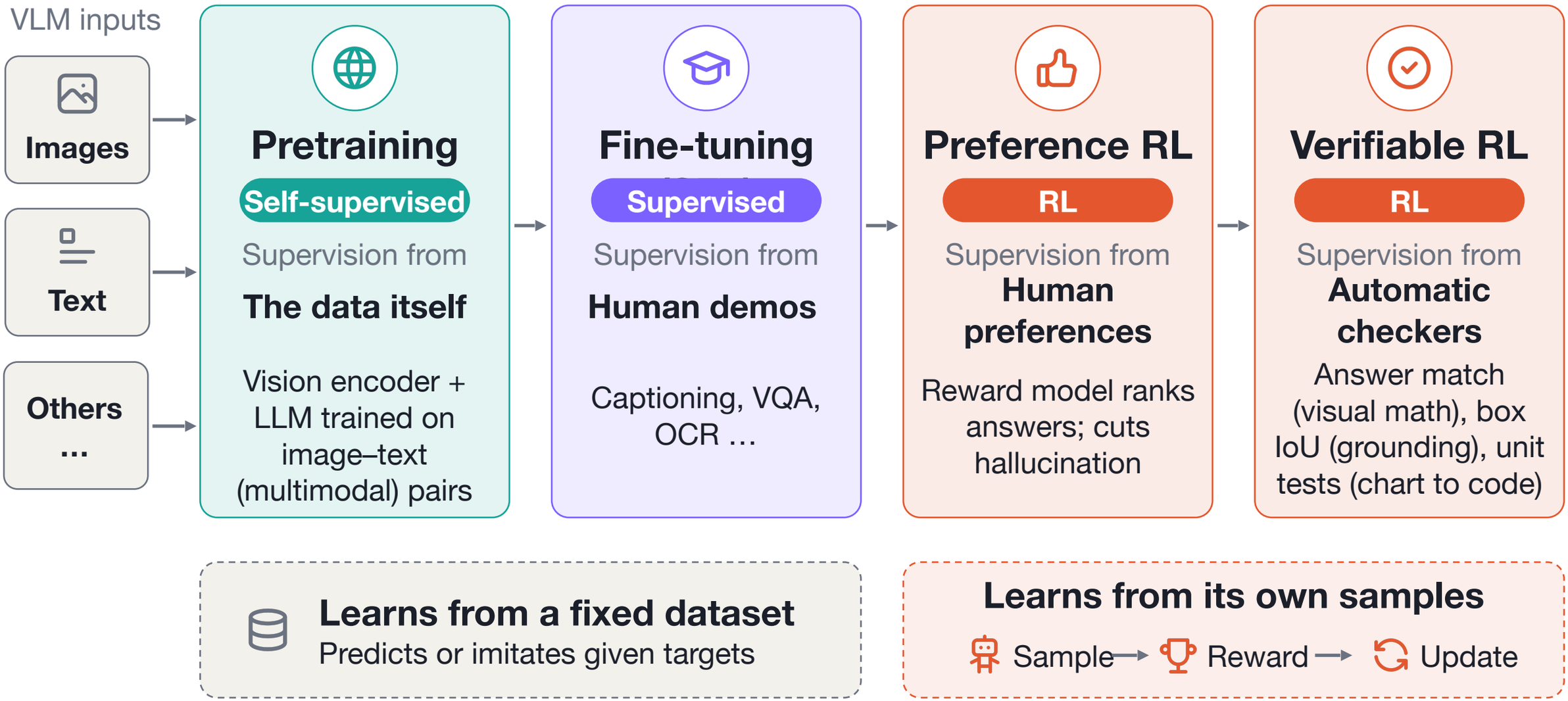
New legal, ethic challenges (e.g., all AI apps inherit bias from a few generic models)

Foundation models

- Key ingredients of *Frontier models* (which are not “new”):
  - **Self-supervised learning**
  - **Reinforcement Learning**
- and new ingredients that were not available a few years ago:
  - Improved hardware
  - **Transformer-based architectures**
  - Availability of data

Foundation models

Frontier models (VLMs) use every form of supervision



Foundation models

- **BERT** (Bidirectional **Encoder** Representations from Transformers)
  - NLP, Google 2018 (currently applied in English-language queries at Google)
  - **transformer** language model with a variable number of encoder layers and self-attention heads
  - trained from unlabeled data from the **BooksCorpus**[4] - 800M words and English **Wikipedia** - 2,500M words.
  - pretrained on two tasks: ***language modeling*** (some tokens masked; trained to predict them from context) and ***next sentence prediction*** (predict if certain sentence was probable or not given the first).

Foundation models

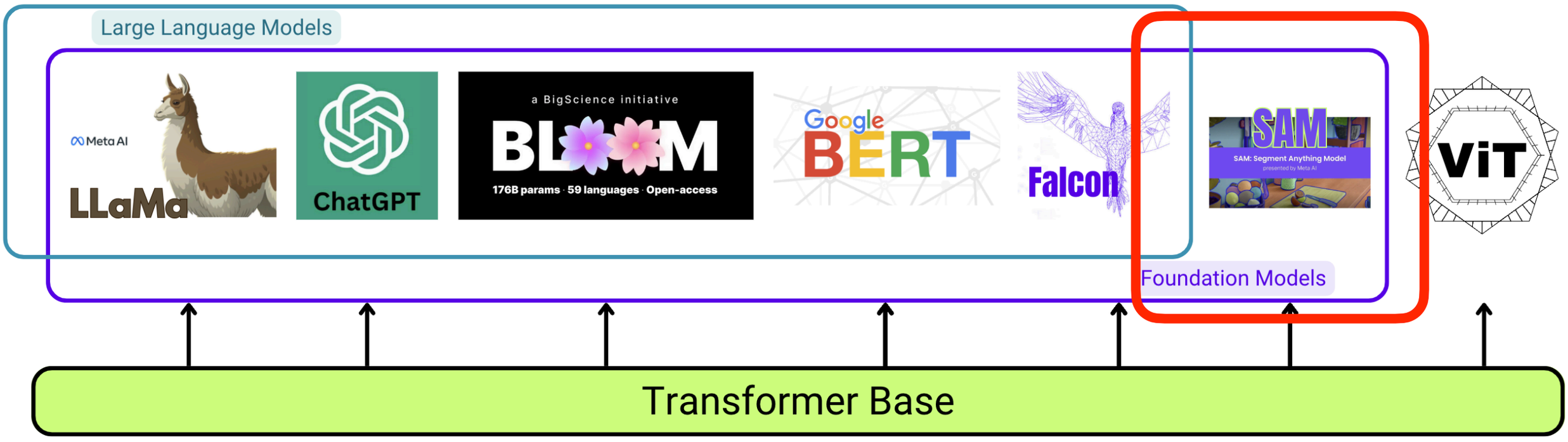
- **GPT family**: *Generative Pre-trained Transformer*
  - NLP, openAI. GPT3 2020; GPT4 March 2023. Generate human-like text
  - **GPT3**: **transformer** decoder network with 2048 token long context and 175 billion params (800 Gb of storage). Largest at the time.
    - trained on ***common crawl*** *(60%), webtext, books, wikipedia (3%)* to *predict what the next token* is
  - >=**GPT4**: pre-training using both public data and "data licensed" from other parties. Then, fine-tuned with reinforcement learning feedback from humans and AI for human alignment and policy compliance. (tech details not public **EVEN MORE!**
    - Also takes **image inputs**

Foundation models

- **GEMINI** family - Google DeepMind - Native multi-modal training
  - Transformer decoder based models. Mixture-of-Experts (MoE) architecture.
  - Main source of **multimodal training**: Web docs, code, books, images, video, audio (filtered for quality/safety)
  - **Post-training:**
    - “supervised fine-tuning” with Human-generated prompts & responses
    - Final alignment with human preference rankings (with RL)
- *Frontier* models (also GPT, Claude, DeepSeek…) are focusing on advanced reasoning ("thinking" models), "vibe coding," and generative interfaces.

Foundation models

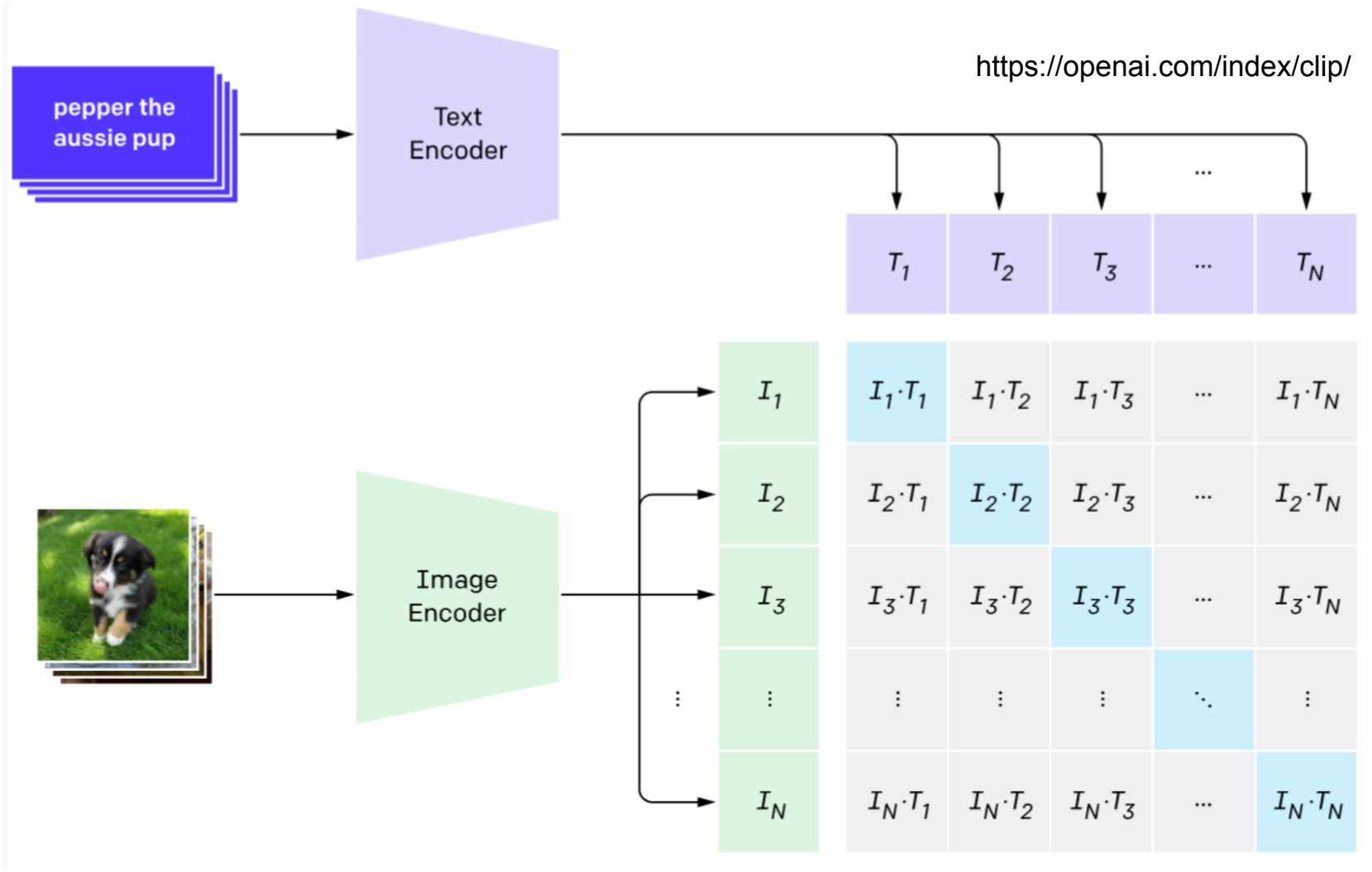
- **Other more specific foundation models**



Foundation models: Large Vision Models

CLIP: Contrastive Language-Image Pre-training

- (Encoder network) Trained on image-text pairs to learn a shared embedding for images and text

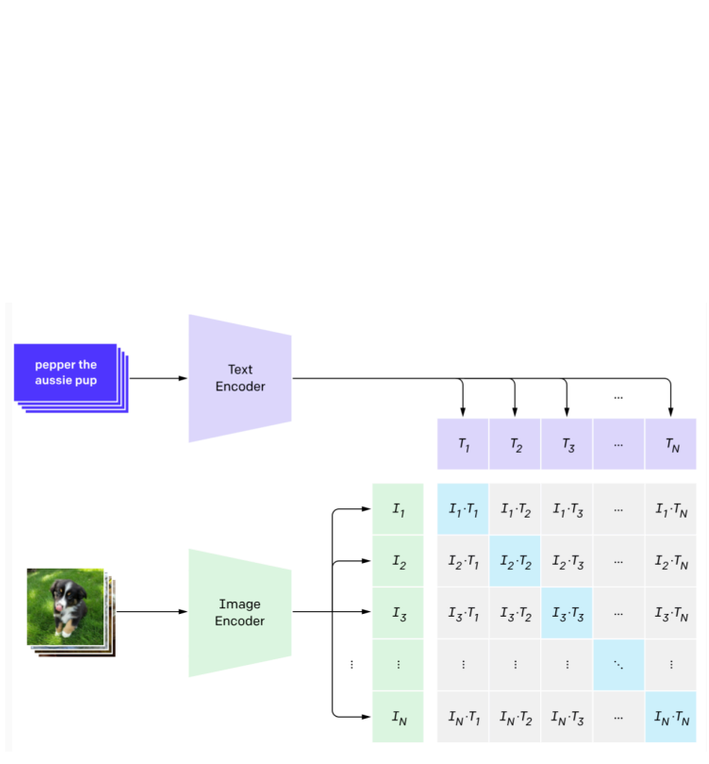


Radford, Alec, et al. "Learning transferable visual models from natural language supervision." International conference on machine learning. PMLR, 2021.

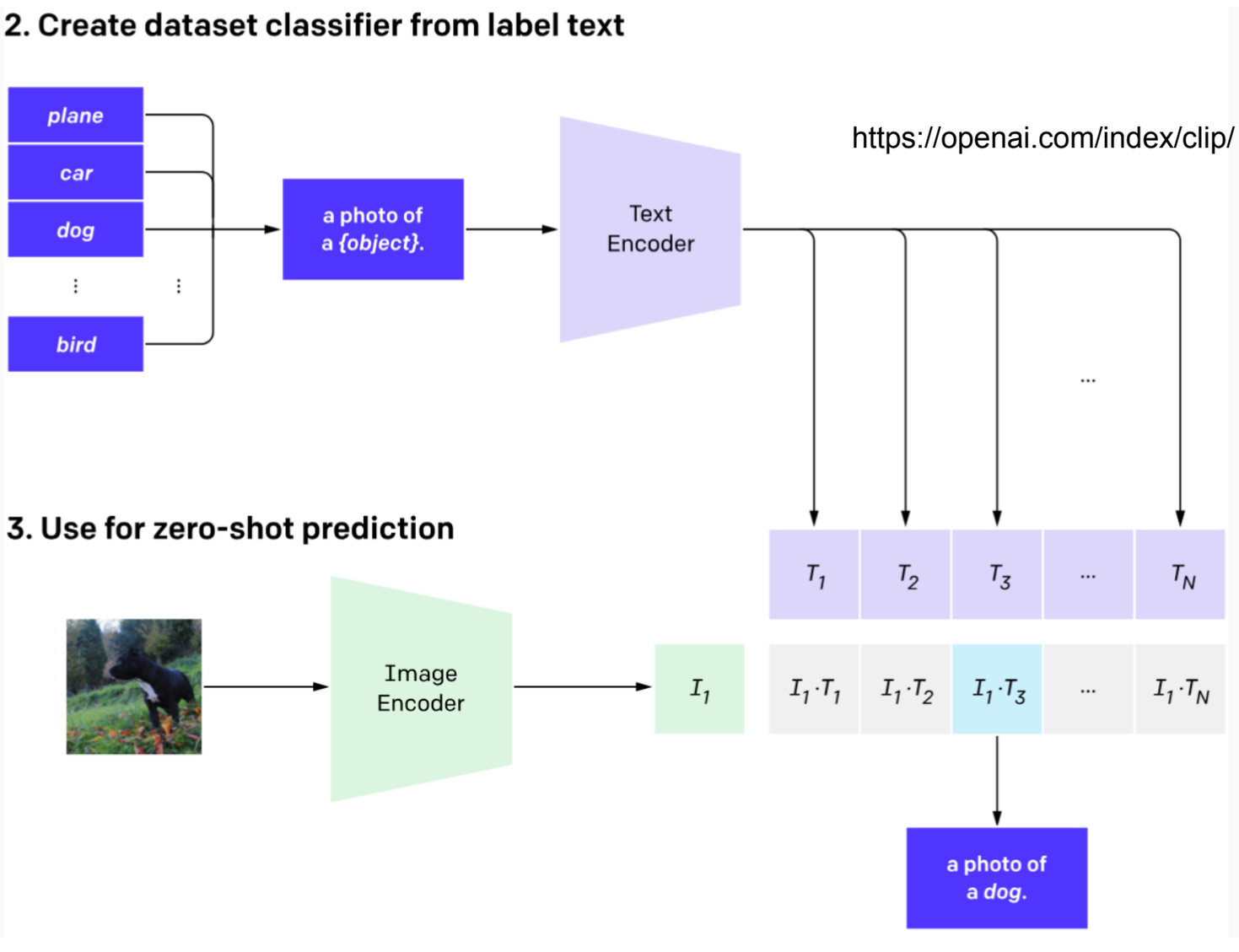
Foundation models: Large Vision Models

CLIP: Contrastive Language-Image Pre-training

- Trained on image-text pairs to learn a shared embedding for images and text
- Tasks like zero-shot image classification & image generation from text







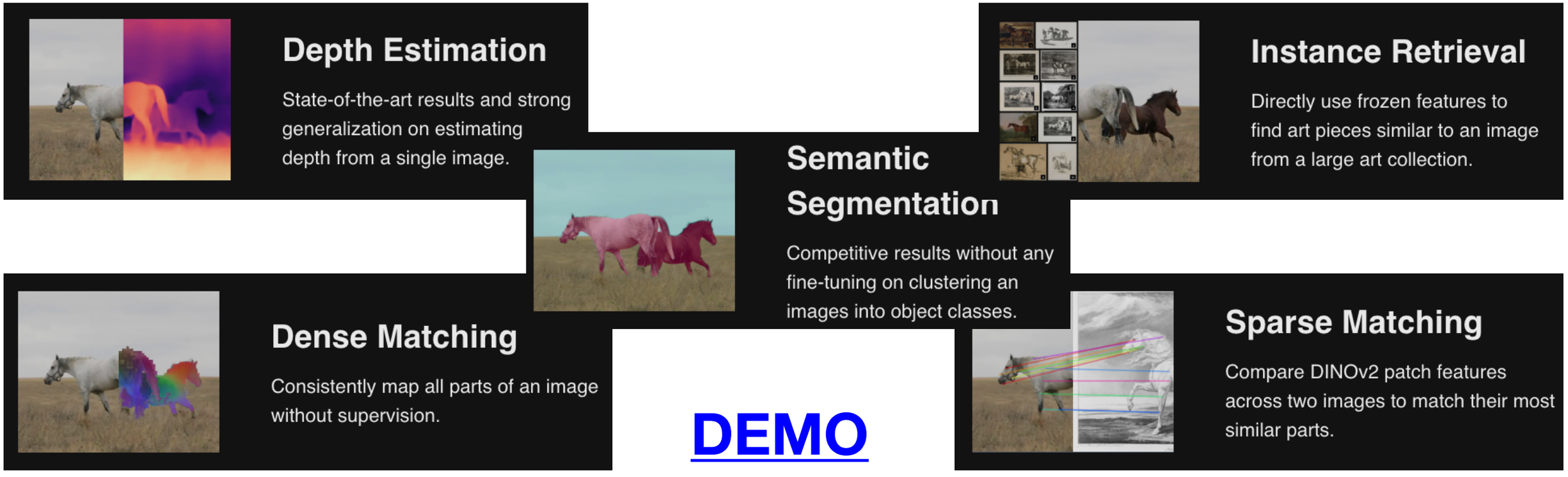
Radford, Alec, et al. "Learning transferable visual models from natural l

Foundation models: Large Vision Models

- *DINOv2: self-distillation with NO labels v2*

  Meta

  - **Self-supervised** learning to obtain **visual representations**
  - Used to train an encoder: teacher-student framework trained on different views of the same image (no labels, no text description used)
  - **Universal** for all kinds of tasks, both image-level & pixel-level, no fine-tunning



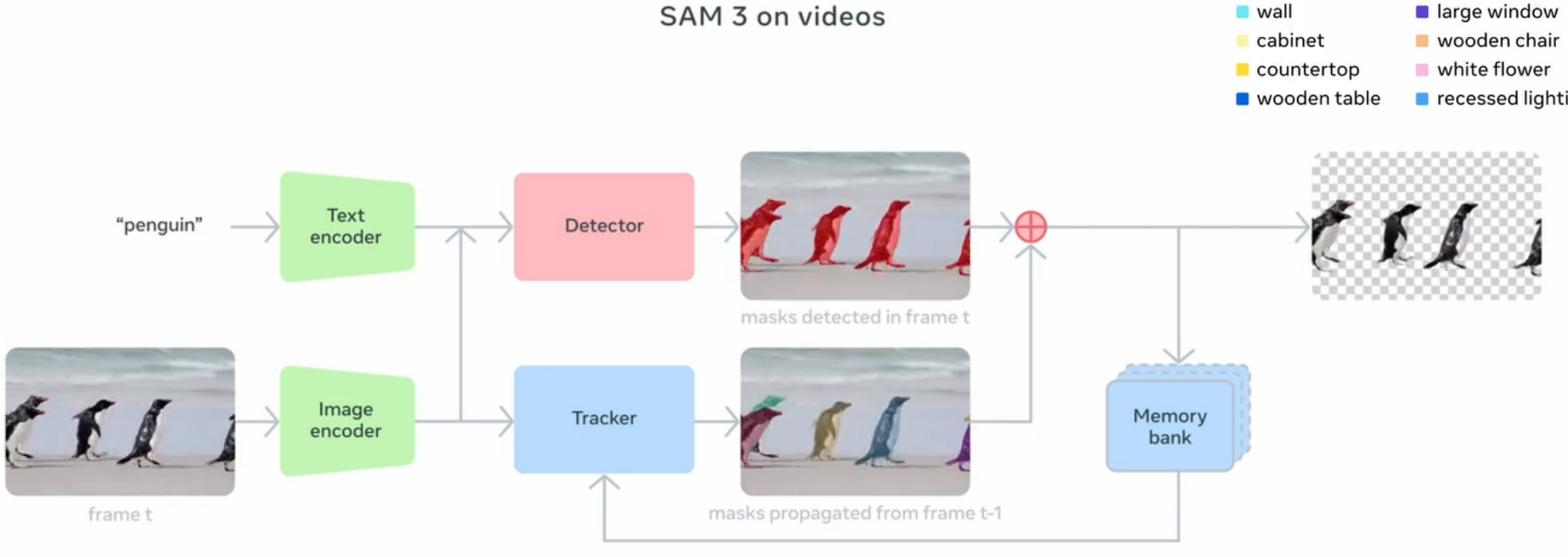
Oquab, Maxime, et al. "DINOv2: Learning Robust Visual Features without Supervision." Transactions on Machine Learning Research Journal (2024)

Foundation models: Large Vision Models

- *SAM: Segment Anything Model*
  - Multiple versions
  - SAM3 includes LLMs and 3D models

DEMO





Meta

DL & *Foundation* models

Image generation?

Foundation models & image generation

NLP + generative image model. **Images from "*prompts*"**

Some of the earliest alternatives:

- **DALL-E** (openAI 2021/22. Based on GPT-3)
- **Stable diffusion** (Stability AI & CompVis LMU 2022. Public code and model. **Diffusion model**: VAE + U-Net + text encoder.)
- **Midjourney** (Midjourney team, 2022)

Generative models: diffusion

NEXT and optional courses

Foundation models & image generation

LLM + diffusion model. **Images from "*prompts*"**

Two self-supervised generative objectives

Language model
Predicts the next token (discrete)

Diffusion model
Denoises a latent step by step (continuous)

LLM conditions diffusion

Separate models

Text prompt

LLM / text encoder → embeddings

Diffusion transformer (denoiser)

VAE decoder → image

e.g. Stable Diffusion 3, Imagen, Qwen-Image

Post-training. Preference optimization and RL now also tune image generators, like for text.

DL & *Foundation* models

Efficiency? Explainability?
Fairness?

Optional courses

Next

- Other supervision strategies:
  - **Unsupervised**
  - (Deep) Reinforcement Learning
- **Generative models**

Bibliography - Resources related to this block

- Stanford classes on deep learning for Computer Vision (http://cs231n.stanford.edu) and Deep Learning (https://cs230.stanford.edu/)

- Berkeley course - CS 198-126: Deep Learning for Visual Data (https://ml-berkeley.notion.site/CS-198-126-Deep-Learning-for-Visual-Data-c77ce2526cb4460b8b02dc56a86bc426)

- Ian Goodfellow, Yoshua Bengio, Aaron Courville, Deep Learning, MIT Press, 2016. http://www.deeplearningbook.org

- *Computer Vision: Algorithms and Applications*. 2nd Edition. Richard Szeliski. https://szeliski.org/Book/

- *Probabilistic Machine Learning: An introduction*. K. Murphy. MIT Press, 2022, probml.ai